TECHNOLOGY

The Brochure Series of Architectural Illustration, Vol. 01, No. 07, July, 1895 Italian Wrought Iron

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THE BROCHURE SERIES

OF ARCHITECTURAL ILLUSTRATION.

VOL. I. JULY, 1895. No. 7.

ITALIAN WROUGHT IRON.

The wrought iron of the middle ages, and of the time of the Renaissance, and even down to the last century, in Italy, France, and Germany showed, in the crudest examples, the principal virtues of all true decorative art. The reason is not far to seek. The difficulties in the way of working the material with ease imposed certain limitations in design and execution which could not well be disregarded. The lack of machinery (which is responsible for much of the uninteresting character of our modern work) necessarily compelled the use of comparatively simple and straightforward methods. It was difficult to avoid the tell-tale marks of the smith's work, and there were limits beyond which his skill could not carry it. Furthermore the designer, taking these limitations into account, learned to make the most of his possibilities, and to adapt his design to the material--to design in the material. How different from the methods generally in use now! Designs made to imitate something done in another material, turned out by the hundred from a machine which leaves no indication of its work, with all interest of craftsmanship lacking, except in places where it may be vulgarly thrown in your face to make it look as if it had been "hand-made."

Clever imitations of old work are produced, and indeed some of the examples shown in our plates are reproductions and not originals; and if we cannot have new designs of equal excellence this is the next most desirable thing. And so far as the illustrations are concerned the difference between the original and the reproduction could never be distinguished.

The subjects chosen for the illustrations of this number are lanterns and torch-bearers. The lanterns were in reality torch-bearers, as they were made for holding masses of combustible material which were held in place by the central spike.

The curious lanterns that decorate the Strozzi Palace at Florence, and of which similar specimens are still attached to the angles of the Riccardi Palace, once the famous residence of the Medici, in the same city, are among, the best examples of their kind still remaining. We are informed by Vasari that these "_lumière miravigliosi_" were the work of one Nicolo Grosso Caparra, a celebrated artificer of the time, by whom it is not unlikely that many of the beautiful rings and cressets which still decorate the old palaces of Siena may have been executed. On the centre

spike was fixed a little iron barrel, containing tow and pitch, while on each of the other spikes a torch was fastened. In some of the old engravings of the festivities given at night by the Grand Dukes of Tuscany, the representations of the effect of this mode of illumination may be seen. It is said that the privilege of affixing such cressets to a residence was one conferred by the State only on the most distinguished citizens, as a peculiar honor, in acknowledgment of services performed.

[Illustration: XLIX. Wrought Iron Lantern on the Strozzi Palace, Florence.]

XLIX,

WROUGHT IRON LANTERN ON THE STROZZI PALACE, FLORENCE.

This is one of the finest examples of Italian wrought iron work still existing, and has frequently served as a model for lanterns in recent times.

[Illustration: L. Wrought Iron Lantern on the Palazzo Guadagni, Florence.]

L.

WROUGHT IRON LANTERN ON THE PALAZZO GUADAGNI, FLORENCE.

[Illustration: LI. Wrought Iron Lantern on the Palazzo Bocella, Lucca.]

LI.

WROUGHT IRON LANTERN ON THE PALAZZO BROCELLA, LUCCA.

[Illustration: LII. Wrought Iron Lantern on the Palazzo Baroni nel Fillungo, Lucca.]

LII.

WROUGHT IRON LANTERN ON THE PALAZZO BARONI NEL FILLUNGO, LUCCA.

[Illustration: LIII. Wrought Iron Torch Holder, Public Square, Siena.]

LIII. WROUGHT IRON TORCH-BEARER, SIENA.

This is attached to a column which bears the group representing Romulus and Remus, and which is situated in the public square near the cathedral.

[Illustration: LIV. Wrought Iron Torch Holder, Siena.]

[Illustration: LV. Wrought Iron Torch Holder, Siena.]

LIV and LV. WROUGHT IRON TORCH-BEARERS, SIENA.

These two plates represent the same torch-holder, viewed from front and side.

[Illustration: LVI. Wrought Iron Torch Holder, Siena.]

LVI. WROUGHT IRON TORCH-BEARER, SIENA.

This together with the preceding example are reproductions of old work.

Reproduction of Architects' Drawings.

The development of photographic processes for the reproduction of drawings which has taken place within the past few years has led to a remarkable increase in the publication of architects' designs, both in the technical journals and in the popular magazines and daily press. Undoubtedly the recent progress of architectural design in America is largely due to the opportunity for comparison thus placed within the reach of architects and draughtsmen who could not otherwise place their productions beside those of their fellows. So important has this become that an architectural paper is now usually judged almost entirely upon the quality of its illustrations, the text matter being not only secondary, but in some cases serving only as a vehicle for the plates. In fact, some of the most valuable and most highly esteemed architectural publications are entirely devoid of text.

It naturally happens that many of the drawings made in the ordinary course of an architect's work sooner or later fall into the hands of the publishers of some of the architectural papers or are required for publication in other directions. When such drawings have been made without a proper knowledge of the requirements of the reproductive processes the result is frequently very unsatisfactory, and in many cases gives an entirely unfair impression of the design, while this difficulty might have been easily avoided by a little forethought, and without any additional labor.

A few fundamental points which can always be kept in mind will enable draughtsmen to make sure that their work will reproduce well, that is to say, will give a fairly truthful reproduction of the original drawing.

There are at present in use a large number of printing processes depending upon photography as a basis, by which drawings may be duplicated, but they can be roughly divided into two main classes according to the character of the original drawings. In general, line drawings may be treated by one process, while those in which there is a gradation of tones or tints, no matter in what way produced (except by distinct lines), require another and entirely different process. Line reproductions may be made in several different ways, but the requirements in the original drawing are the same in each. The first requisite is that a drawing shall be made in absolutely black ink on white paper, and with clear, firm lines. With a little care it is just as easy to make a drawing in this way as any other, and a satisfactory reproduction can be assured when it is kept in mind that nothing but black will give the best results. In the early days of process work it was customary to use India ink ground by the draughtsman, but excellent liquid inks, such, for instance, as that made by Charles M. Higgins & Co., have taken the place of this, at a great saving of labor and trouble. It is only necessary to take care that the ink is new and not too watery, and that a sufficient amount is carried in the pen to insure a black line. Gray lines, although full and continuous, are very apt to be ragged and broken in the reproduction. Aside from this first condition there are few others which are really mandatory. A drawing made with vigorous, well-defined lines and rather open in treatment will, as a rule, make the most satisfactory reproduction.

There is never difficulty in getting a good reproduction from such work as that by which Mr. H.P. Kirby or Mr. D.A. Gregg is known. For this purpose their style could hardly be improved upon. A drawing can be made with fine and delicate lines and still reproduce well if there is not too much difference in size between the original and the reproduction required. In general, the best results can be obtained by making the plate about two thirds the size of the original.

Drawings in colored inks on tinted paper are difficult to reproduce satisfactorily, and of all combinations a bluish ink upon a yellowish paper is to be avoided.

In general, it can be said that everything, even including line drawings in pen and ink, _can_ be reproduced by the half-tone processes, the quality of the plate depending upon the character of the original. Water-colors, monochrome drawings in wash, pencil drawings and any combinations of these, are reproducible, but with varying success. The same conditions which apply to line work also hold good to a considerable extent in the present case. A combination of vigorous black ink lines and lighter more delicate work put in with thinned or gray ink will in all probability be very unsatisfactory, as the chances of holding the relation between the two, or in fact of preserving the lighter lines at all, without over-emphasizing the darker portions, will not be very great. Delicate drawings can seldom be reproduced without giving a background tint all over, and this usually destroys the life and snap of the original. This is especially true of drawings upon reddish or yellowish paper, which on this account should be avoided if possible. It should be borne in mind that yellow and red photograph dark; and blue, light. This often makes a great difference of effect in the reproduction and sometimes makes it impossible to get satisfactory results at all, especially in delicate drawings.

Pencil drawings made with light lines will not reproduce well, as there is too little contrast in color between the lines and the paper; but sketches made with a soft pencil and strong contrasts frequently give surprisingly good results.

When drawings are to be made, especially for reproduction, the question of expense is often of importance. Plates made from pen drawings now cost about ten cents a square inch, while half-tone plates made of metal for printing on an ordinary printing press with type matter cost about twenty-five cents a square inch.

By using specially prepared process papers, which, if not sold by a local dealer in artists' materials, can be had of Messrs. Wadsworth, Rowland & Co., or Frost & Adams, drawings can be made in pencil or black crayon which can be reproduced by the cheaper process, and will give excellent results. Considering the ease with which this work can be done and the satisfactory results obtained, it is surprising that it has not been more generally adopted. The only drawback to working upon this paper is the fact that no erasures or changes can be made without ruining the surface of the paper.

In connection with what has already been considered in relation to the reproduction of drawings, it may be well to refer to the making of plates from photographs. The selection of a good photograph is of the first importance. It should be brilliant, and with all the contrast of light and shade and as much detail as

possible, for something is always lost in both these respects in the process of reproduction. A good plate can be made from a good photograph, but cannot from a bad one. The process is the same as that referred to above for the reproduction of wash drawings, etc., and the cost the same, about twenty-five cents a square inch. The half-tone plates in THE BROCHURE SERIES, made by The Blanchard & Watts Engraving Company, Boston, are good examples of first-class work of this description.

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No subject at present occupies so important a place in the thoughts of American architects as that of architectural education, if the space given to it in recent publications is an indication of its importance. The proceedings of the annual convention of the American Institute of Architects, held last autumn in New York, have just been published, and no less than five papers are included which deal with one or another phase of this subject. The later numbers of the professional journals also contain several noteworthy contributions to the discussion. Mr. Barr Ferree's criticism in _The Architectural Review,_ of the methods of training pursued in the School of Fine Arts in Paris, have led to several papers by adherents of the French system and to a well-considered editorial in the same paper. But the most important contribution to the question is that of Mr. Henry Rutgers Marshall in the last number of _The Architectural Record,_ which also contains a descriptive article upon the Royal Polytechnicum at Berlin and its course of study.

There is very little in any of these articles which adds to the existing knowledge on this largely discussed subject; it is what might be considered a rethreshing of old straw, and the main value of all of the articles is in the presentation, which may appeal to readers who have not before thought of the matter in all of its bearings. The papers read before the convention begin with the report of the committee on education, by Mr. Henry Van Brunt. In this Mr. Van Brunt advocates the careful and systematic study of architectural history; and it was the purpose of the report to bring out discussion which might lead to valuable suggestions to the architectural schools

upon the study of this subject. Mr. Geo. B. Post, of New York, Professor Ware, of Columbia College, and several others took part in the discussion which resulted in merely recommitting the question to the committee on education, as it was not considered advisable to take any definite action which would bind the Institute to a settled policy on this question. Mr. Louis H. Sullivan, of Chicago, in a thoughtful paper complained that education stifles and kills the spirit of modern architectural work, and that the natural and spontaneous love for beauty found in all human beings gives place, under our modern systems of instruction, to the dry formalities of reproducing old and dead styles.

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Mr. Frank Miles Day and Mr. R.W. Gibson, each in his own way, described the advantages of foreign travel and the best methods to pursue, as well as the most important ends to be attained.

Mr. Russell Sturgis, in a scholarly paper upon the subject of Greek architecture applied to modern buildings, gave a number of precepts for the proper use of Greek forms and methods of building as applied to our modern conditions. He closed his article with a lot of receipts much in the style of an architectural cook book, for the application of his theories concerning Greek architecture.

These articles, as will be seen from the above indications, dealt in most cases with one phase only of architectural education. They are all of course important in their way, as contributing to the general discussion of the subject, but each in turn gives only a partial view.

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Mr. Marshall, however, starts with the intention of making a full and fair statement of existing conditions and logically draws his inferences as to the best methods of meeting them. He has the valuable qualification of being able to consider his subject judicially and of writing excellent and readable English, as has already been pointed out in these columns, in the review of his recently published book, "Æsthetic Principles."

He divides the subject for convenience into three main headings: first, the use of tools, including in his classification the executive function of handling men as well as tools in the narrower and legitimate sense; second, the nature of materials employed; and third, the general principles of beauty. Under the first heading Mr. Marshall makes one important suggestion, which is at variance with common practice among architects. After pointing out the importance of studying design in the solid, that is, constantly keeping in mind that the forms which are to be designed have three dimensions, and that a geometrical projection, such as a plan or elevation, only partially represents its appearance, he advocates the more general use of perspective drawing in designing. By this is not meant the making of pretty sketches after the design is all determined, to mislead impressionable clients, but the serious study of a design to determine its appearance from different points of view. In fact his suggestion is that the usual order of proceeding shall be reversed, and that the design shall be made in perspective and then translated into accurate terms of such

geometrical projections as are needed to guide the practical workman in carrying out the work.

In treating of materials Mr. Marshall takes up separately the materials of construction and the materials of design, meaning by the latter the vocabulary in which the architect expresses his ideas, or the accumulation of architectural forms making up the various historic styles, so-called. He emphasizes the importance--in which point he agrees with all the other writers above referred to--of a wide and catholic knowledge of architectural history and a careful study of all styles.

In summing up in the portion of the article devoted to the general principles of beauty as applied to architecture he gives a clear and concise statement of the reasons why beauty is in itself a necessary and desirable element in architecture, and roughly analyzes the conditions under which it exists.

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Brochure Series Competition, No. I.

The judges in the BROCHURE SERIES COMPETITION No. I, for a Piano Case have awarded the three prizes as follows:--First Prize, $25.00, to Mr. A.B. Le Boutillier of Boston; Second Prize, $15.00, to Mr. Edward F. Maher of Boston; Third Prize, $10.00 to Mr. James C. Green of Brooklyn, N.Y.

The report of the judges with reproductions of the prize designs will be given in the next issue of the BROCHURE SERIES.

The drawings have been retained by the H.F. Miller Piano Co. for exhibition at their warerooms and will be returned by them at the close of the exhibition.

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Clark Medal Competition.

The seventh annual competition for the Robert Clark Testimonial, under the auspices of the Chicago Architectural Club, is herewith presented.

CONDITIONS.--The competition is open to architectural draughtsmen under thirty years of age, residents of the United States, and not practicing architects.

The author of each design must execute all drawings without assistance, and non-adherence to these conditions will cause the rejection of the design or designs in question.

The awards will be made by the adjudicating committee on the "Robert Clark Testimonial Competition," and are: First prize, gold medal; second prize, silver medal; third prize, bronze medal.

The two designs receiving honorable mention will receive special bronze medals.

The three prize drawings shall become the property of the Chicago Architectural Club.

AN ART SCHOOL.--A gentleman wishing to share his large and valuable collection of paintings, statuary and architectural fragments with his townsmen, has decided to

place them in a building which he proposes to erect for the study of architecture, painting and sculpture.

The building is to face the town square, and is to be not more than one hundred and fifty feet in its greatest dimensions.

It shall consist of one story and a high basement.

The first story shall contain the following rooms:

1st. A large entrance gallery for the placing and hanging of statuary and paintings. This hall should be the main feature of the plan, and should be carefully arranged for convenient and advantageous display, without destroying the architectural effect. It may be one continuous hall or divided into parts, at the discretion of the architect. It may be lighted from above.

2d. A large glass-covered court to contain architectural fragments.

3d. An amphitheatre, to seat about two hundred, for lectures on art subjects. A library and an assembly hall.

4th. Four class rooms. These rooms should be well-lighted and of easy access to the court and gallery.

5th. A janitor's room and an office for the custodian. These rooms may be small, but should be conveniently placed either at the entrance to the building or to the grounds.

As the number of the students is limited, the size of the rooms is of less importance than the circulation, convenience and artistic beauty of the whole.

The building, being the home of the arts, should be pure in style and classical in feeling, though not necessarily archæological.

Drawings required, viz.: One plan and one section at the scale of one-sixteenth of an inch to one foot, and the front elevation at the scale of one-eighth of an inch to one foot.

Drawings to be rendered at will; to be mounted on strainers 28x40, without frames or glass.

A sealed envelope containing the name and full address of the author, with place and date of birth, must be securely fastened to each drawing; the drawings and envelopes themselves must not be marked by a device of any kind.

Drawings must be delivered to John Robert Dillon, secretary, Chicago Architectural Club, at the club house, 274 Michigan Avenue, Chicago, on or before Friday, November 15, 1895, charges to be prepaid. All drawings not receiving prizes will be returned at the expense of the contributor.

L.J. MILLET, Chairman; R.C. SPENCER, JR. IRVING K. POND, The Adjudicating Committee on the Robert Clark Testimonial.

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Personal.

The death of Richard Morris Hunt, at Newport, R.I., on July 31, deprives the architectural profession in this country of the man who, since the death of Mr. Richardson, has been its most distinguished representative. His influence upon American architecture is possibly less directly traceable than that of Richardson, and was more of a personal nature through association with his brother architects, while Richardson's example was his most important legacy to the profession.

Mr. W.S. Hebbard will, on September 1, occupy new offices in the Grant Building, San Diego, Cal., which he is just completing for U.S. Grant, Jr., Esq.

Among the recent additions to the working force of Mr. Aiken, Supervising Architect of the Treasury, are Mr. F.B. Wheaton, formerly with Messrs. Longfellow, Alden, & Harlow, and Mr. Rice, formerly with Wheelwright & Haven.

Mr. Geo. H. Ingraham, who has recently opened an office at 6 Beacon Street, Boston, is now absent on a short European trip.

Mr. George E. Barton, for several years with Cram, Wentworth, & Goodhue, of Boston, has just started for a tour of England and France, with the special purpose of studying the domestic and church architecture of the smaller cities and towns.

Mr. C.H. Alden, who has lately returned from six months' travel, mostly in Italy, has made a careful study of the brick and terra-cotta architecture of Northern Italy. He has just entered the office of Messrs. Wyatt and Nölting, Baltimore.

Each year since the University of Pennsylvania Traveling Scholarship was founded, a prominent member of the T Square Club has been the winner; and that Mr. Percy Ash, ex-president of this club, should carry off the prize this year is particularly gratifying.

Mr. Ash has twice before competed, and each time came out a close second; but his old luck did not entirely forsake him, for in his venture for the Roman Scholarship Prize he was very near to the front, winning honorable mention.

H.L. Duhring, Jr., was a close second for the U. of P. Scholarship.

At the last regular T Square Club meeting, but two sets of drawings were submitted. The program called for a "Garden for a Palatial Country House," and required a plan of the house and terrace at 1/8" scale, and a plan and section of the entire garden at a scale of 1/32 of an inch.

The problem was modeled after the _projet_ given at the Ecole des Beaux-Arts, and required so vast an amount of graded wash work in color, as to intimidate many of the regular competitors. A.C. Muñoz, who won first mention, submitted three drawings, two of them nearly three by four feet, while Albert Kelsey was disqualified for not having fulfilled the requirements by omiting the 1/8" scale plan.

Some discussion advocating the postponement of the competition took place, but Kelsey seemed to prefer being disqualified rather than further exert himself; and possibly the knowledge that three draughtsmen in Day's office and two in Cope &

Stewardson's office had two unfinished designs to complete, may have influenced him. In spite of the result of this competition the eleven points previously won by Mr. Kelsey give him the highest average for the past year's work.

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Notes.

Of the many minor or industrial arts which enter into a complete architectural production, that of the smith is one of the most fascinating, and strangely enough, it is one which at the present time has the fewest workers who can be worthily compared with those of the past. In the estimation of many of the most prominent and exacting architects of the country there is but one maker of ornamental wrought iron in America who can be trusted to intelligently carry out the spirit of a fine design. Why this should be so it is hard to say, but the fact remains that most of the best iron work done in this country in recent years has come from the shop of John Williams of New York; and architects, it may be said, instinctively turn to him for work of this class.

The characteristics which distinctively belong to the art of the smith, the limitations of material and the purpose for which the finished work is intended are all taken into account and each element given its due importance. To Mr. H.B. Stillman, associated with Mr. Williams, who has for a number of years taken personal charge of this branch of the business, is largely due the success which has attended the efforts of his house.

[Illustration]

The suburban house architecture of the towns about Boston is of exceptional interest and its quality is generally considered to be equal, if not superior, to that of any other locality in the country. The reason for its superiority in design and consequent interest is largely traceable to the influence of such architects as Peabody & Stearns, Winslow & Wetherell, Andrews, Jaques & Rantoul, Hartwell & Richardson and a number of others who have given especial attention to residence work.

One of the most attractive collections of houses of this class which we have seen is contained in a finely printed little booklet issued by Dexter Bros., of Boston. It contains photographic illustrations of eleven houses designed by the architects named above, and others. The houses themselves are hardly more attractive than the excellently chosen and finely reproduced photographic views. Messrs. Dexter Bros., upon application, will send this booklet to any architect or draughtsman.

For fastening any sort of work to stone or brick the clever expansion bolt, patented and manufactured by Isaac Church, of Toledo, is, on every count, the best device to specify. Patterns for every special use imaginable are made by him and fully described in his catalogue.